

Introduction to Credit Securities

IIT Madras, 17 April 2026

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Two ways to raise capital. Two ways to invest money.



Debt

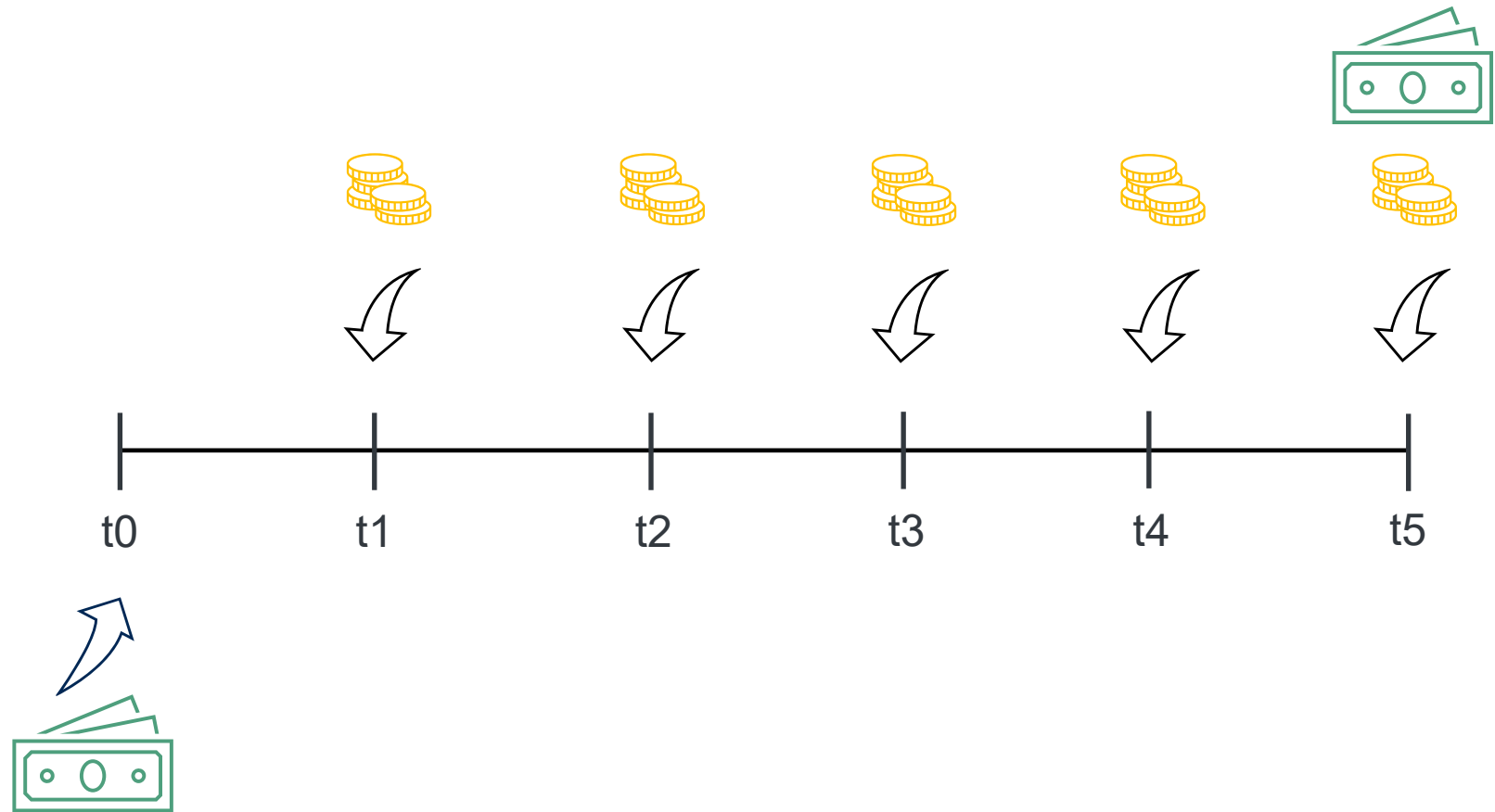
Bonds & loans
Lower risk
Lower expected return



Equity

Stocks (shares)
Higher risk
Higher expected return

How does debt work?



Cashflows from a simple fixed coupon bond

Fixed income securities are financial instruments that pay investors:

- Fixed interest payments until their maturity date
- The remaining principal amount at maturity



Features of bonds

- **Issuer:** corporations, sovereign governments, non-sovereign/quasi government entities, supranational entities
- **Maturity date:** varies from 1 day to 30 years or more.
 - Money market securities: bonds with maturity ≤ 1 year
 - Capital market securities: bonds with maturity > 1 year
 - Perpetual bonds: with no maturity
- **Par/face value:** principal value to be repaid at maturity. If bond is being sold at a price $>$ par value, it is said to be sold at a premium. If the selling price $<$ par value, it is called a discount bond.
- **Coupon rate and frequency:** coupon rate is the annual percentage of par value that will be paid to the bondholders. Bonds can make annual, semi-annual, quarterly or monthly coupon payments. Bonds which do not make any coupon payments are called zero-coupon or pure discount bonds.
 - Fixed coupon bonds
 - Floating coupon bonds/Floating rate notes: interest depends on current market rate of interest/reference rate, e.g., LIBOR. Often a margin is added to the reference rate.
- **Currency:** in which payments will be made.

Why do people buy bonds?



Predictable Income

Investors purchase bonds to receive regular interest payments, providing a **predictable stream of income**. This appeals to those seeking **stability and consistent returns**.

Bonds are less risky than stocks due to fixed payments and principal return at maturity. They help **diversify portfolios** and are ideal for **conservative investors** focused on **capital protection**.



Lower Risk & Diversification

Bond markets



Primary Market

- Issuers sell bonds directly to investors for the first time.
- Funds raised support projects or operations.
- Investors receive newly issued bonds.

Secondary Market

- Investors buy and sell bonds in the secondary market after issuance.
- Provides **liquidity** and allows investors to enter or exit positions at market prices.

Trading Methods

- Bonds are usually traded **over-the-counter (OTC)** rather than on exchanges.
- Banks such as JPMorgan Chase are **market makers**.

Time value of money

Money now is worth more than money later

What's the future worth of today's \$100?

Let's say the yearly interest rate is 5%

\$100 today becomes:

- After 1 year: $100 + (5\% \text{ of } 100) = 100 + 5 = \105
- After 2 years: $105 + (5\% \text{ of } 105) = 105 + 5.25 = \110.25
- After 3 years: $110.25 + (5\% \text{ of } 110.25) = 110.25 + 5.5125 = \115.7625
- .
- .
- .
- After n years: $100 \times (1 + 0.05)^n$

What's \$100 on 17th April 2027 worth today?

Let's say the yearly interest rate is 5%

\$A today becomes $A \times (1+0.05)^3$ in 3 years.

Solve the equation: $A \times (1+0.05)^3 = \$100$

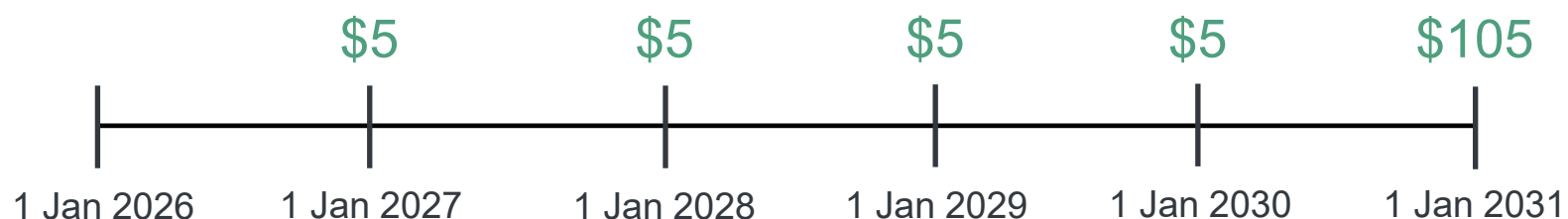
We get $A \approx \$86.38$

This means that \$100 in 2027 is worth only \$86.38 today

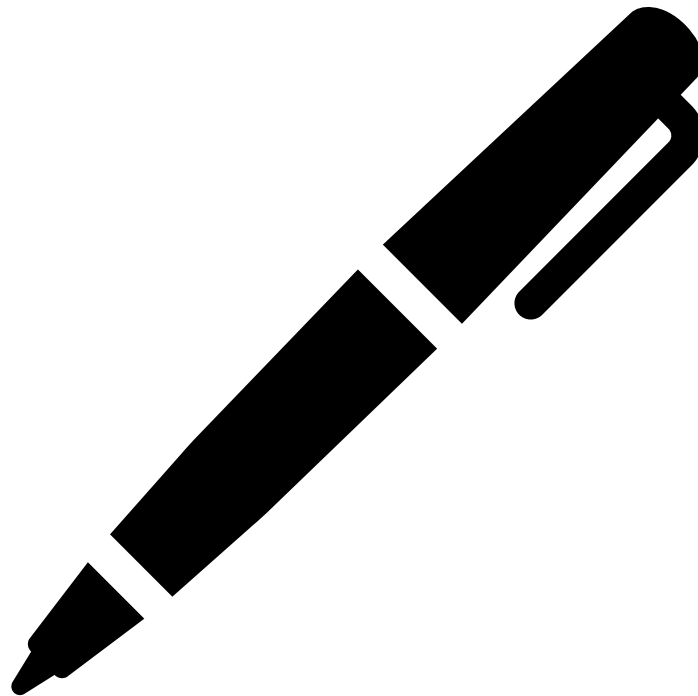
Cashflow discounting

To price any instrument, we simply need to add up the present value (PV) of each of its future cashflows.

$$PV = \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \dots + \frac{CF_n}{(1+r)^N}$$



$$PV = \frac{5}{(1.04)} + \frac{5}{(1.04)^2} + \frac{5}{(1.04)^3} + \frac{5}{(1.04)^4} + \frac{105}{(1.04)^5}$$



Invisible hand of the market

Person 1 thinks interest rates will **vary** over next 3 years as follows:

- 2026-27: 5%
- 2027-28: 6%
- 2028-29: 7%

Person 1 is willing to pay **\$83.96** today for a \$100 zero coupon bond with maturity 3 years

Person 2 thinks interest rates will remain **constant** at 5% over next 3 years

Person 2 is willing to pay **\$86.38** today for a \$100 zero coupon bond with maturity 3 years

Market price will be determined by what market participants think about how interest rates are going to evolve in the future

Bond prices and interest rates are inversely related

Today, 17th April 2026: The government issues a 1-year maturity \$100 zero-coupon bond when the prevailing interest rate is **4%**. Let's call this **issue A**

Price at which you can buy it is $100/(1.04) = \mathbf{\$96.15}$

Next Friday, 24th April 2026: The prevailing interest rate moves up to **5%** ! The government again issues a 1-year maturity \$100 zero-coupon bond. Let's call this **issue B**

Price at which you can buy it is $100/(1.05) = \mathbf{\$95.24}$

After next Friday, nobody would want to buy issue A for \$96.15 and get \$100 after one year when they could buy issue B for \$95.24 and get \$100 after one year.

What happens then?

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What happens then?

There is higher supply of issue A as compared to demand, which makes the price of issue A fall till it eventually matches the same return as issue B.

Bond prices and interest rates are inversely related

If prevailing interest rates go up, the prices of already issued bonds (that were issued in a low interest rate environment) fall to match the ROI of newly issued bonds.

Conversely, if prevailing interest rates were to drop, the prices of already issued bonds would rise.

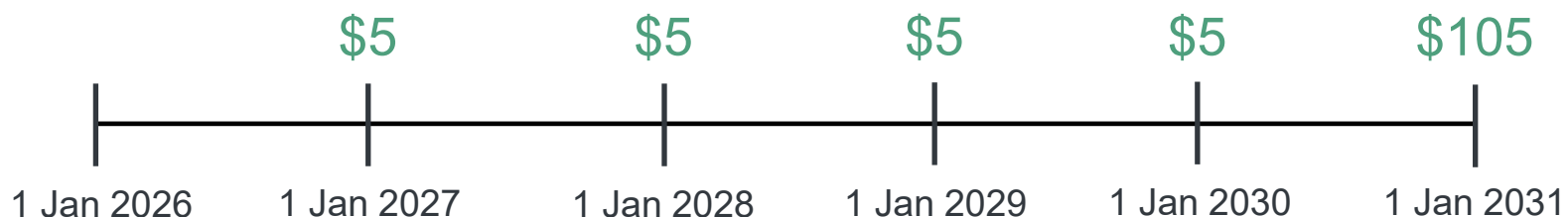
All coupon bonds are portfolios of ZCBs

This sequence of cashflows can be:

Portfolio 1: Five ZCBs of \$5 notional and maturities ranging from 1 through 5 years, plus one ZCB of \$100 notional and 5-year maturity

OR

Portfolio 2: One 5% coupon bond with annual coupon payments, \$100 notional and 5-year maturity



Efficient markets imply no arbitrage opportunities.

In an efficient market, the prices of both these portfolios should be equal, otherwise there will be **arbitrage!**

Yield to maturity (YTM)

If spot rates for borrowing over next T years are $z_1, z_2, z_3, \dots, z_T$, then the price of a bond with notional N , maturity T years and annual coupon rate q can be expressed as:

$$PV = \frac{qN}{(1 + z_1)} + \frac{qN}{(1 + z_2)^2} + \frac{qN}{(1 + z_3)^3} + \dots + \frac{qN + N}{(1 + z_T)^T}$$

$$PV = \frac{qN}{(1 + y)} + \frac{qN}{(1 + y)^2} + \frac{qN}{(1 + y)^3} + \dots + \frac{qN + N}{(1 + y)^T}$$

Here, y is the **yield-to-maturity**

It is the rate by which if you discounted all the future cashflows of the bond, you would arrive at the current market price.

It is also the total annualized rate of return expected on a bond if held until maturity, assuming all coupon payments are reinvested at the same rate.

Discount, par and premium

If bond price < notional **discount** bond yield > coupon rate

If bond price = notional **par** bond yield = coupon rate

If bond price > notional **premium** bond yield < coupon rate

Yield curve

A plot showing how bond yields vary across different maturity dates

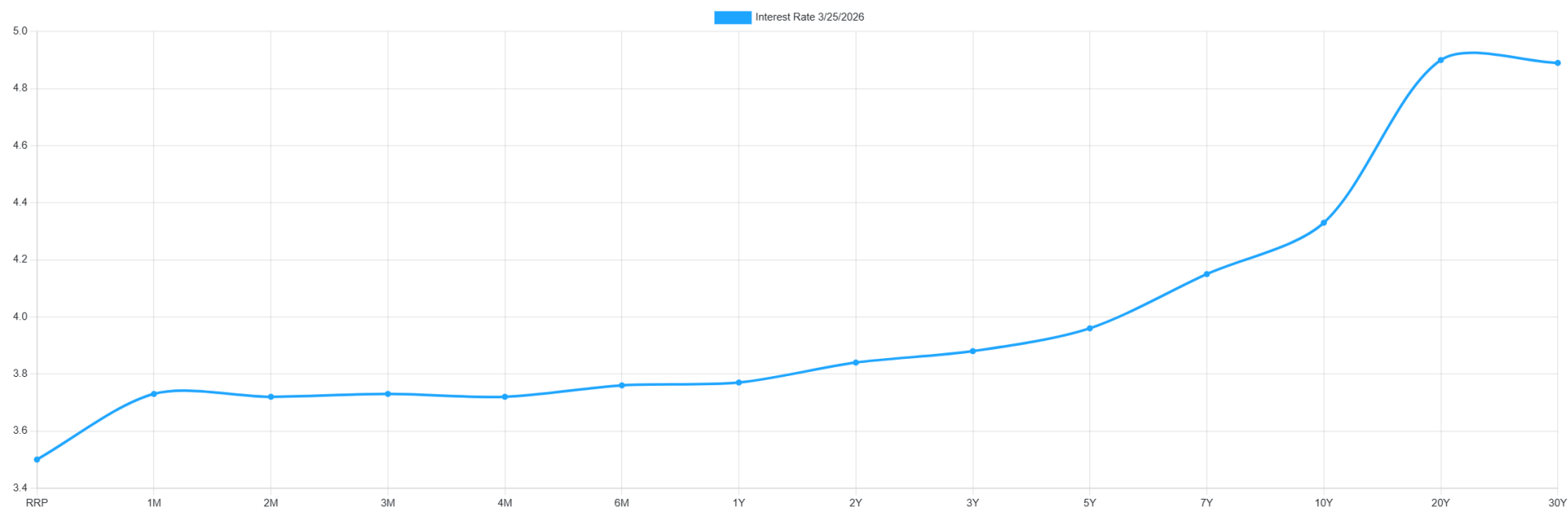
Date ↑	1 Mo ↑	1.5 Mo ↑	2 Mo ↑	3 Mo ↑	4 Mo ↑	6 Mo ↑	1 Yr ↑	2 Yr ↑	3 Yr ↑	5 Yr ↑	7 Yr ↑	10 Yr ↑	20 Yr ↑	30 Yr ↑
03/02/2026	3.74	3.74	3.74	3.72	3.68	3.68	3.54	3.47	3.49	3.62	3.82	4.05	4.64	4.70
03/03/2026	3.75	3.75	3.74	3.71	3.67	3.68	3.55	3.51	3.50	3.63	3.83	4.06	4.65	4.70
03/04/2026	3.75	3.75	3.75	3.71	3.68	3.68	3.58	3.54	3.55	3.67	3.87	4.09	4.67	4.72
03/05/2026	3.75	3.74	3.72	3.70	3.68	3.68	3.59	3.57	3.59	3.72	3.92	4.13	4.71	4.74
03/06/2026	3.75	3.74	3.72	3.69	3.67	3.66	3.55	3.56	3.59	3.72	3.93	4.15	4.74	4.77
03/09/2026	3.75	3.73	3.72	3.71	3.68	3.68	3.56	3.56	3.58	3.71	3.90	4.12	4.70	4.72
03/10/2026	3.75	3.74	3.71	3.71	3.69	3.68	3.56	3.57	3.58	3.73	3.93	4.15	4.74	4.78
03/11/2026	3.75	3.73	3.70	3.71	3.69	3.68	3.60	3.64	3.64	3.79	3.98	4.21	4.82	4.86
03/12/2026	3.76	3.74	3.72	3.72	3.69	3.70	3.66	3.76	3.75	3.88	4.06	4.27	4.86	4.88
03/13/2026	3.75	3.74	3.71	3.72	3.69	3.70	3.66	3.73	3.74	3.87	4.07	4.28	4.89	4.90
03/16/2026	3.75	3.73	3.72	3.72	3.69	3.72	3.64	3.68	3.69	3.80	4.00	4.23	4.83	4.86
03/17/2026	3.74	3.72	3.72	3.72	3.69	3.71	3.63	3.68	3.68	3.79	3.98	4.20	4.81	4.85
03/18/2026	3.73	3.72	3.72	3.73	3.71	3.74	3.68	3.76	3.76	3.87	4.05	4.26	4.84	4.88
03/19/2026	3.73	3.71	3.72	3.73	3.71	3.76	3.73	3.79	3.79	3.88	4.06	4.25	4.82	4.83
03/20/2026	3.73	3.71	3.72	3.74	3.73	3.79	3.80	3.88	3.90	4.01	4.20	4.39	4.97	4.96
03/23/2026	3.73	3.71	3.72	3.74	3.72	3.77	3.76	3.83	3.85	3.95	4.15	4.34	4.93	4.91
03/24/2026	3.74	3.72	3.72	3.74	3.73	3.78	3.81	3.90	3.93	4.03	4.21	4.39	4.95	4.94
03/25/2026	3.73	3.72	3.72	3.73	3.72	3.76	3.77	3.84	3.88	3.96	4.15	4.33	4.90	4.89

Source: U.S. Department of the Treasury website (home.treasury.gov)

Yield curve

A plot showing how bond yields vary across different maturity dates

The US treasury yield curve below shows the yields of “**on the run**” US treasury bonds of different maturities. On the run = most recently issued bond of a given maturity.



Source: U.S. Department of the Treasury website (home.treasury.gov)

Yield curve

When the US Treasury issues a new bond or note, it sets the coupon rate at auction to be as close as possible to the yield that clears the market. The goal is to have the bond price at or just below par at issuance.

So the coupon on an on-the-run Treasury is essentially a snapshot of where market yields were at the time that particular security was auctioned.

The on-the-run securities across the curve were all issued at different times, when market conditions were different. So they may all have different coupons. For example you might see:

- The on-the-run 2-year bond was auctioned last month with a 4.25% coupon
- The on-the-run 5-year bond was auctioned two months ago with a 3.875% coupon
- The on-the-run 10-year bond was auctioned three months ago with a 4.00% coupon
- The on-the-run 30-year bond was auctioned last quarter with a 4.375% coupon

None of these bonds have the same coupon, and none of them are necessarily trading at par right now, because yields have moved since each was issued.

Zero curve

Curve plotting yields of zero-coupon bonds versus different maturities.

ZCB with **1 year** maturity: price **p1**, yield **z1**

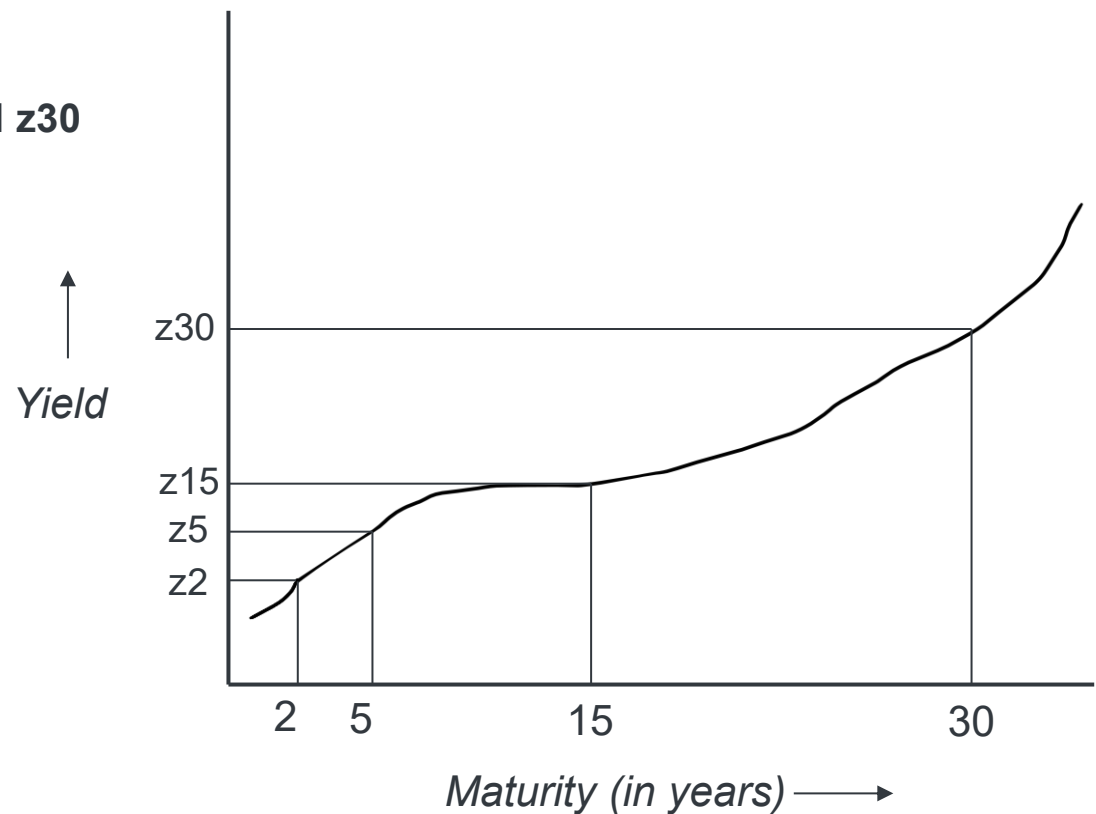
ZCB with **2 year** maturity: price **p2**, yield **z2**

ZCB with **5 year** maturity: price **p5**, yield **z5**

.

ZCB with **30-year** maturity: price **p30**, yield **z30**

$$p_T = \frac{100}{(1 + z_T)^T}$$



Zero curve bootstrapping

Not many zero-coupon bonds are traded in the market, and the ones that are are not very liquid.

Bootstrapping = building the zero-curve using market observable prices of coupon paying US treasuries that are traded more frequently.

P_t = price of coupon bond of maturity t , notional N and coupon rate q

$$P_1 = \frac{N + Nq}{(1 + z_1)^1}$$

$$P_2 = \frac{Nq}{(1 + z_1)^1} + \frac{N + Nq}{(1 + z_2)^2}$$

$$P_3 = \frac{Nq}{(1 + z_1)^1} + \frac{Nq}{(1 + z_2)^2} + \frac{N + Nq}{(1 + z_3)^3}$$

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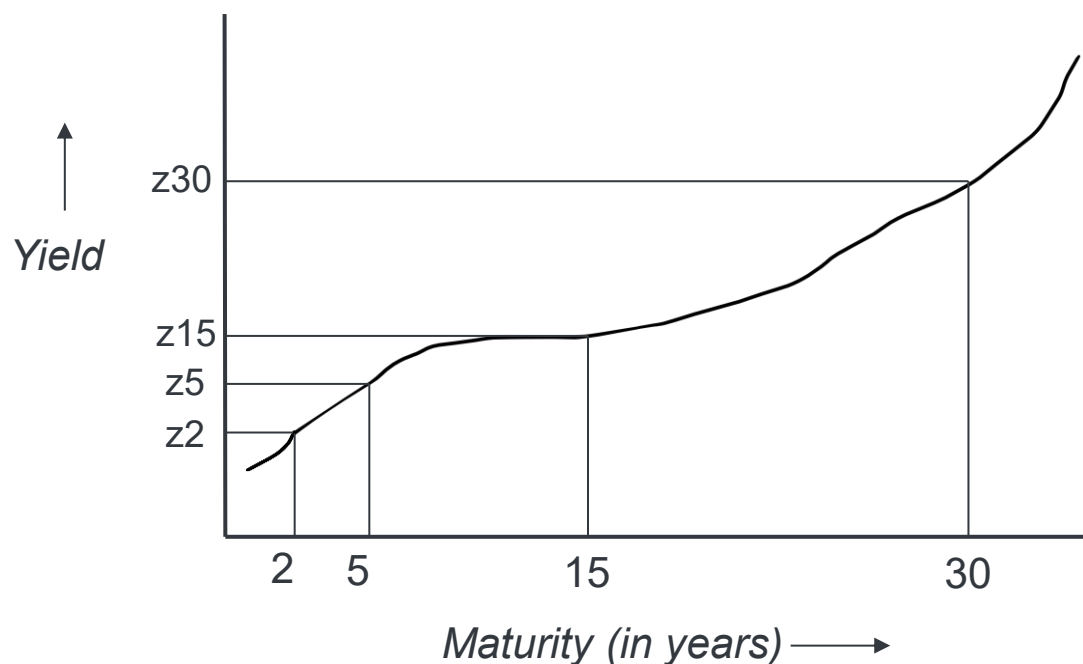
Constants: N, q

Market observables: P_t

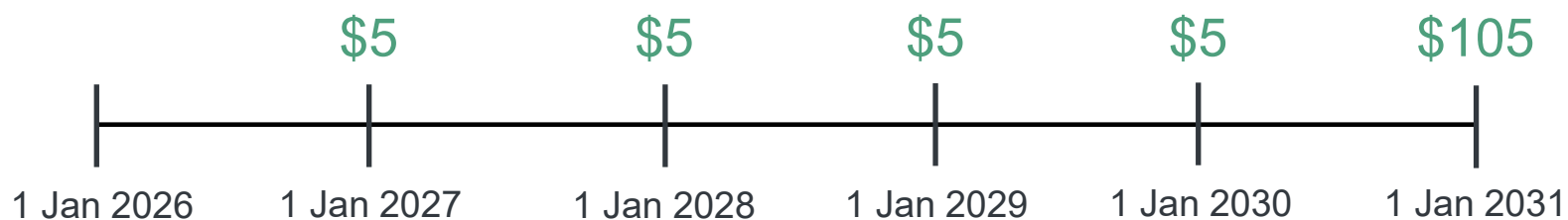
Unknowns: z_t

Solve iteratively

Zero curve can be used for *risk-free discounting* of future cashflows



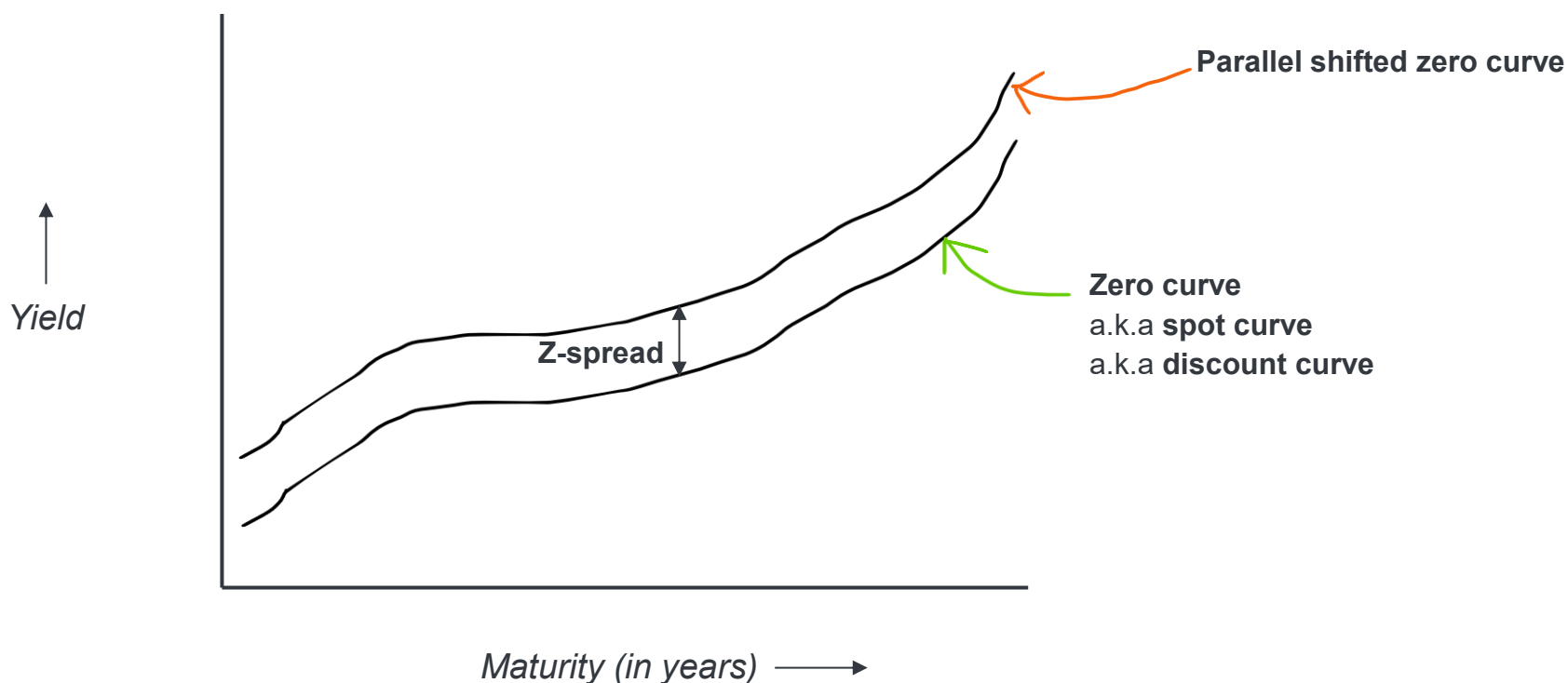
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Corporate bonds are riskier than government bonds

If two bonds have the exact same features except that they are issued by different entities, they could have different prices and yields based on the public perception about their creditworthiness.

Z-spread is the constant spread added to the entire zero curve such that discounting a bond's cashflows using these adjusted rates exactly reproduces its market price.



How to compute Z-spread of a corporate bond

Step 1: Construct the zero curve z_t using market prices of treasury bonds of different maturities

Step 2: Obtain market price p of the corporate bond (say notional N , coupon rate q , maturity T)

Step 3: Solve the following equation for Z :

$$p = \frac{Nq}{(1+z_1+Z)^1} + \frac{Nq}{(1+z_2+Z)^2} + \frac{Nq}{(1+z_3+Z)^3} + \dots + \frac{N+Nq}{(1+z_T+Z)^T}$$

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Bonds with more credit risk typically trade at higher Z-spreads ($\therefore$ lower prices, higher yields).

Think of it like this: the bond holder needs to be compensated with a higher yield because they hold greater risk (default risk, recovery uncertainty, liquidity risk, etc.)

Credit risk – survival based approach

Instead of thinking in terms of spread, we can think in terms of probability of default

$S(t)$ = Prob(no default up to time t)

$Q(t)$ = Prob(default before time t)

$$Q(t) = 1 - S(t)$$

Prob(company defaults between t and $t+\Delta t$ | it did not default till t):

$$\begin{aligned} &= \frac{Q(t + \Delta t) - Q(t)}{S(t)} \\ &= \frac{S(t) - S(t + \Delta t)}{S(t)} \end{aligned}$$

$\lambda(t)$ = hazard rate defined so that is the probability of default between time t and $t+\Delta t$ is $\lambda(t)\Delta t$

Survival curve
(i.e. cumulative survival probability that issuer survives until time t)


$$\begin{aligned} \frac{S(t) - S(t + \Delta t)}{S(t)} &= \lambda(t)\Delta t \\ \frac{dS(t)}{dt} &= -\lambda(t)dt \\ S(t) &= e^{-\int_0^t \lambda(u) du} \end{aligned}$$

Bond price expressed in survival terms

$$P = \sum_t D(t) [S(t) CF_t + (S(t-1) - S(t)) R]$$

The diagram illustrates the components of the bond price formula $P = \sum_t D(t) [S(t) CF_t + (S(t-1) - S(t)) R]$. Arrows point from the following labels to their corresponding terms in the formula:

- Bond price** points to P .
- Risk-free discount factor** points to $D(t)$. Below this label is the formula: $= \frac{1}{(1 + z(t))^t}$.
- Probability of surviving to time t** points to $S(t)$.
- Cashflow at time t** points to CF_t .
- Probability of default in that period** points to $S(t-1) - S(t)$.
- Recovery value in case of default** points to R .

Spreads tell you how much extra return you get for holding more risk.

Survival curves tell you how likely you are to get paid.

Z-Spread is how the market quotes credit risk.

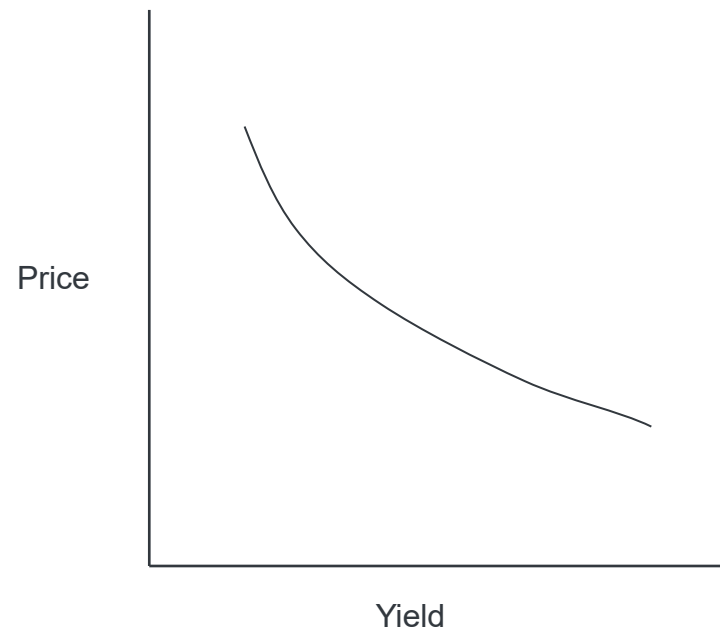
Hazard rates are how we model it.

This framework is then used to price more complex products such as credit default swaps (CDS)

Interest rate risk

As interest rates change, bond prices also change

Interest rate risk is measured by the sensitivity of price to changes in yield



Interest rate risk

Macaulay duration: weighted average time to maturity

$$\begin{aligned}D_m &= \sum_{k=1}^T k \cdot \omega_k \\ \sum_{k=1}^T \omega_k &= 1 \\ \omega_k &= \sum_{k=1}^T \frac{C_k / (1+y)^k}{P} \\ &= \frac{PV(C_k)}{P}\end{aligned}$$

Sensitivity of bond prices to yield changes

$$\begin{aligned}P &= \sum_{k=1}^T \frac{C_k}{(1+y)^k} \\ \frac{dP}{dy} &= \frac{-1}{1+y} \sum_{k=1}^T k \cdot \frac{C_k}{(1+y)^k} \\ \frac{1}{P} \frac{dP}{dy} &= - \frac{D_m}{1+y} \\ &= -D_m^* \text{ (modified duration)}\end{aligned}$$

Bond Ratings



What are bond ratings?

Bond ratings indicate how safe or risky a bond is for investors.

Agencies like **CRISIL, S&P, and Moody's** assign ratings based on the issuer's financial health.



Who assigns ratings?



Why do ratings matter?

Higher-rated bonds are safer; lower-rated bonds offer higher returns.

Investment Grade (IG/HG): Lower risk, lower returns

High Yield (HY): Higher risk, higher returns

Bond Ratings

AAA	Prime
AA+, AA, AA-	High grade
A+, A, A-	Upper medium grade
BBB+, BBB, BBB-	Lower medium grade
BB+, BB, BB-	Non-investment grade speculative
B+, B, B-	Highly speculative
CCC+, CCC, CCC-	Substantial risks
CC	Extremely speculative
C	Default imminent
D	In default